

KINKELDEY & HOMBURG

DATA CITIES

Data Visualization and Storytelling W25/S26

Sarah × Emma

The Beginning

Sarah and Emma

Budget

Costs

Leisure Prices

What's Left

Activities

Free in Hamburg

Conclusion

A HAMBURG STUDENT STORY

**Same City.
Same Activities.
Different Budgets.
Different Possibilities.**

**AFFORDABILITY OF
LEISURE ACTIVITIES FOR
STUDENTS**

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AGENDA

TOPIC & MOTIVATION

STORYLINE

DATA & METHOD

VISUALIZATION DESIGN

AI USE

TOPIC & MOTIVATION

Leisure activities for students in data cities

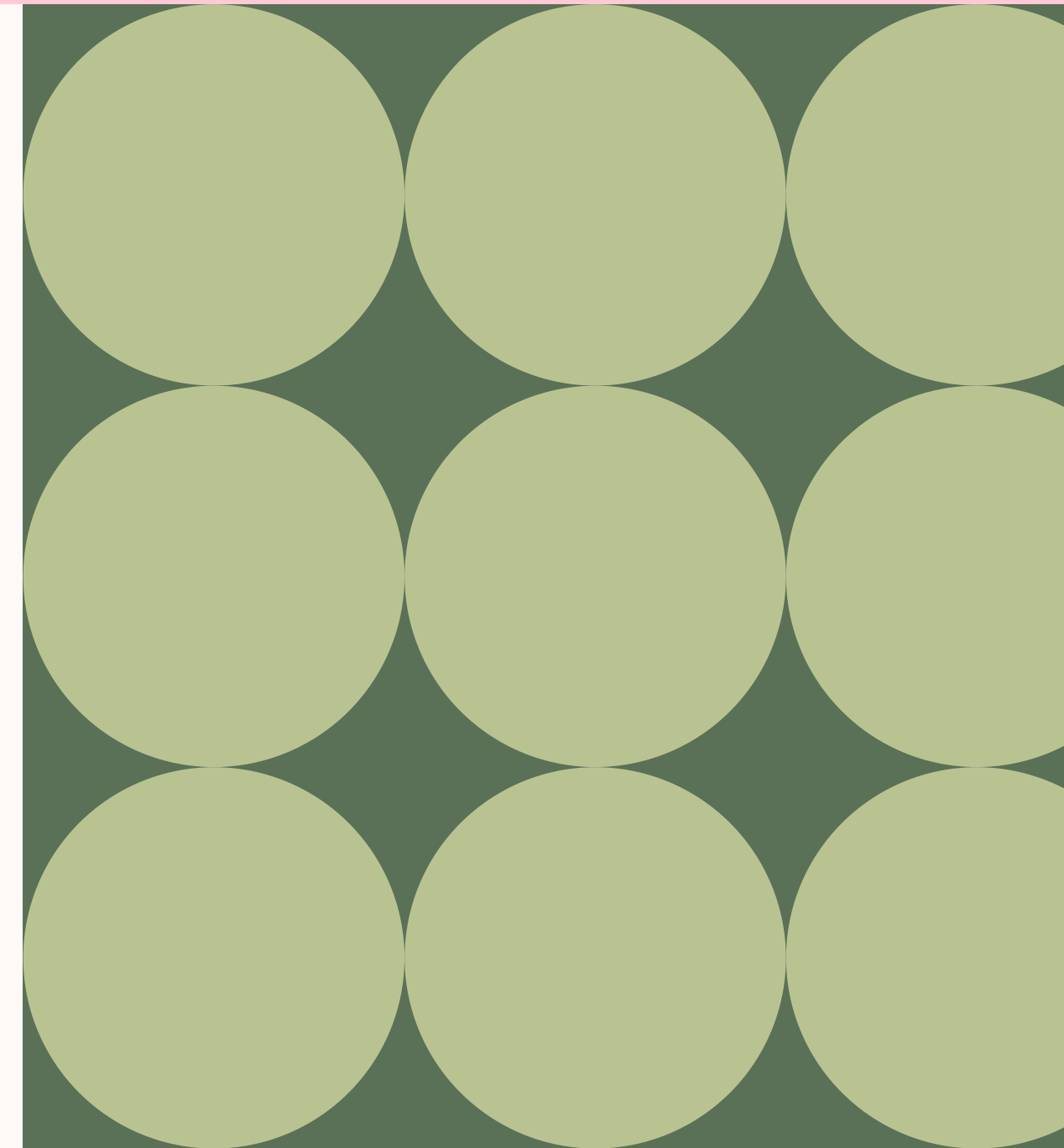
- **TOPIC**

- how affordable are leisure activities for students 2024 vs. 2016
- development
- guide for students

- **WHY**

- we are affected/can relate
- we experience & hear that prices are rising but how does it really affect us
- we want to know: which activities exactly can I still do with my budget left
- helpful to know solutions

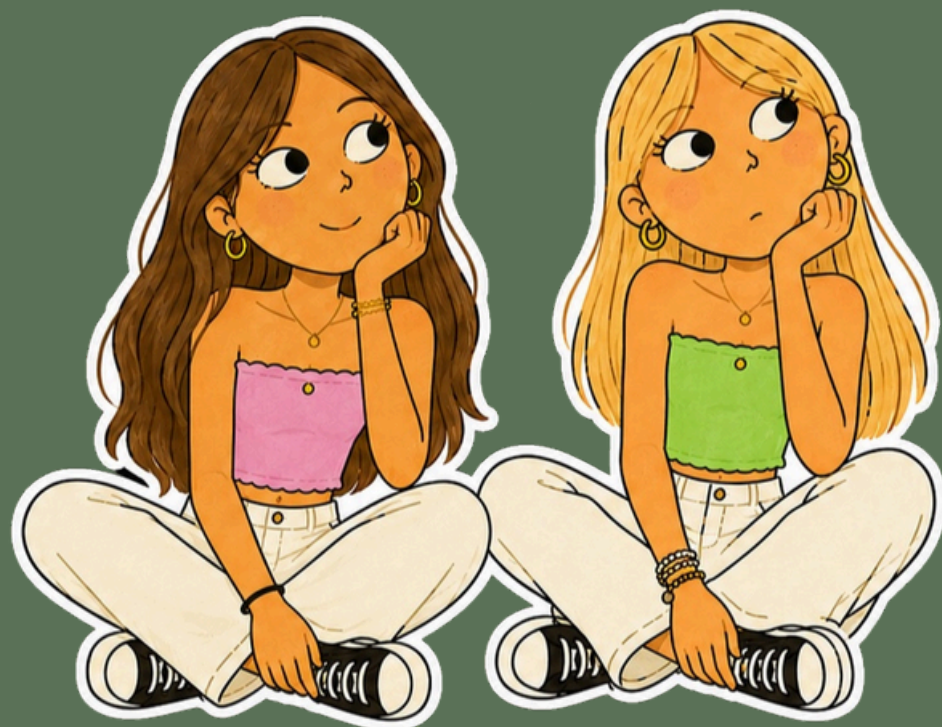
→ We want to show how rising costs in Hamburg affect whether students can still enjoy normal leisure activities



01

Meet Sarah & Emma

- Sarah and Emma are sisters
 - Both studied the same course in Hamburg
 - Sarah studied 2014–2017
 - Emma studied 2021–2024
- **But how affordable was student life then vs. now?**



02

Budget Breakdown

- What money did they have available? (income, BAföG, part-time jobs, family support)
 - What were their fixed monthly expenses? (rent, health insurance, semester fee, groceries, transport, phone)
- **How much money was left after paying the essentials?**

STORYLINE

03

What Can You Actually Afford?

- Compare Sarah's and Emma's remaining budget
- What could each of them afford with the money left?
- How has affordability changed over time?
- Student tips: free and discounted activities, services, and offers in Hamburg.

DATA & METHOD

METHOD

- what data is needed?
- Excel as a data source

SOURCES

income:

- Studierendenbefragung DSW Deutschland (BAföG, Job, Studentjob)

rent:

- Zahlenspiegel Einzelbericht 2016-2024
- Mietpreise Wohnungsmarkt
- WG-Gesucht Mietpreisentwicklung

living costs:

- Studierendenbefragung DSW Deutschland

child benefit:

- Höhe von Kindergeld und Kinderfreibeträgen 1998 - 2025

Inflation increase:

- Verbraucherpreisindex in Hamburg von 2015 bis 2024
- Mitgliedsbeiträge für Fitnessstudios in Deutschland 2025
- Durchschnittlicher Kino-Eintrittspreis bis 2029
- Entwicklung der Verbraucherpreise von alkoholischen Getränken in Deutschland bis 2025
- Preisentwicklung Döner in Hamburg 2016 bis 2024 in €

LIMITATIONS

- mixed data sources and methods
- index only shows change
- very specific leisure activities
- difficulties to find Hamburg based data not just for Germany in general
- finding general Data from different years had to be compiled because only individual reports were available
- Some data had to be requested (wg-gesucht)

VISUALIZATION DESIGN

01 Barchart

- visualize the comparison between both budgets by using bars

Monthly Average Budget of Students

Budget in Euro

Sarah Emma

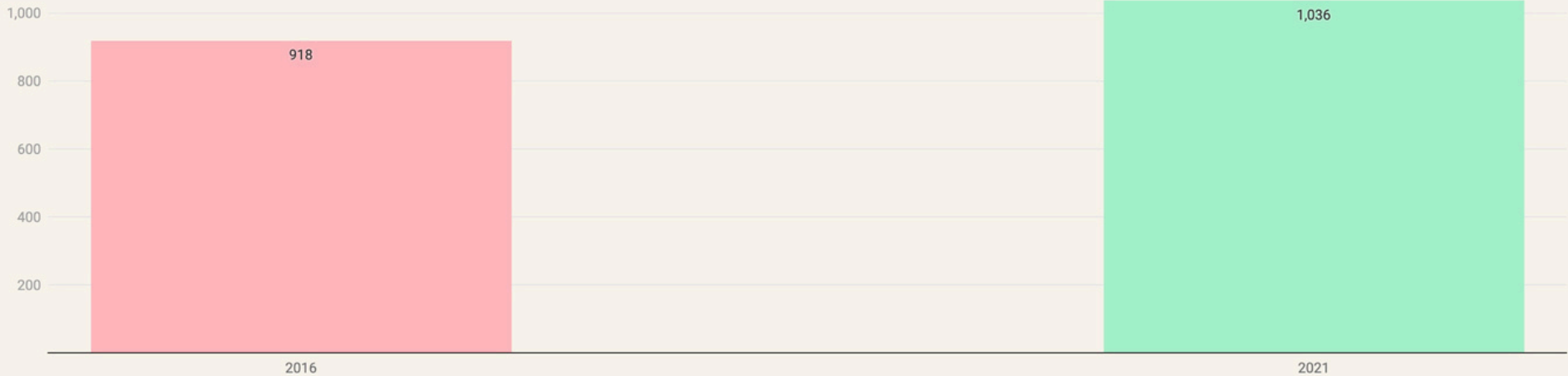


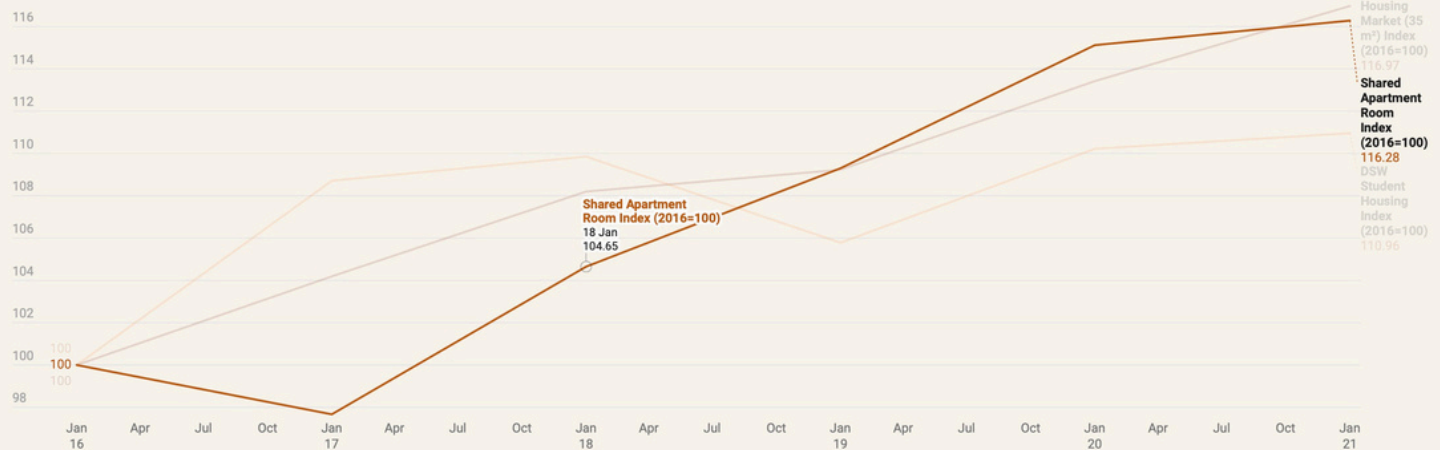
Chart: Leon Hämling • Source: [Studierendenwerk](#) • [Get the data](#) • Created with [Datawrapper](#)

02 Linechart

- visualize the development/trend of rent prices as an index

Hamburg's students are paying more for housing

Indexed development (2016 = 100) of student housing, WG rooms, and apartment rents in Hamburg.

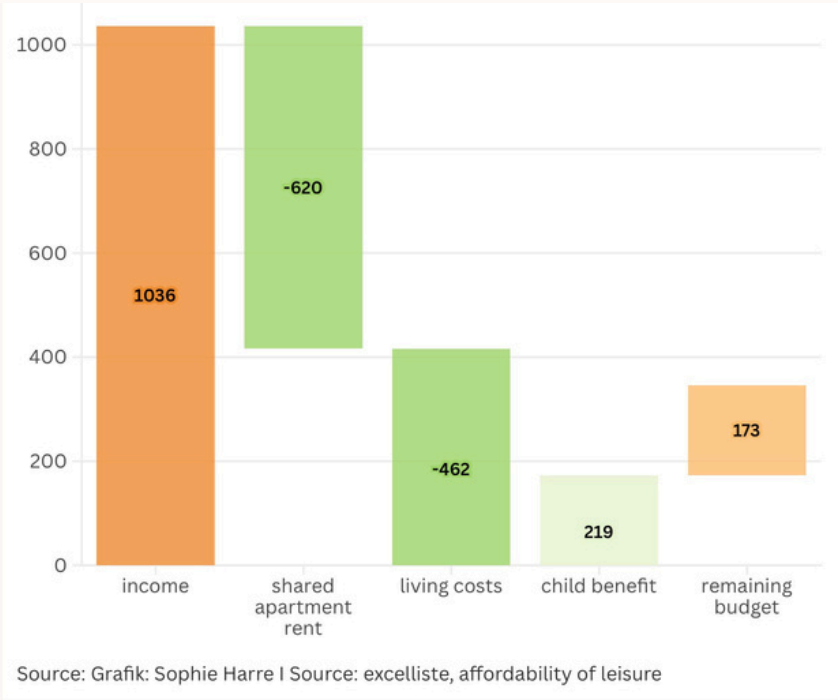
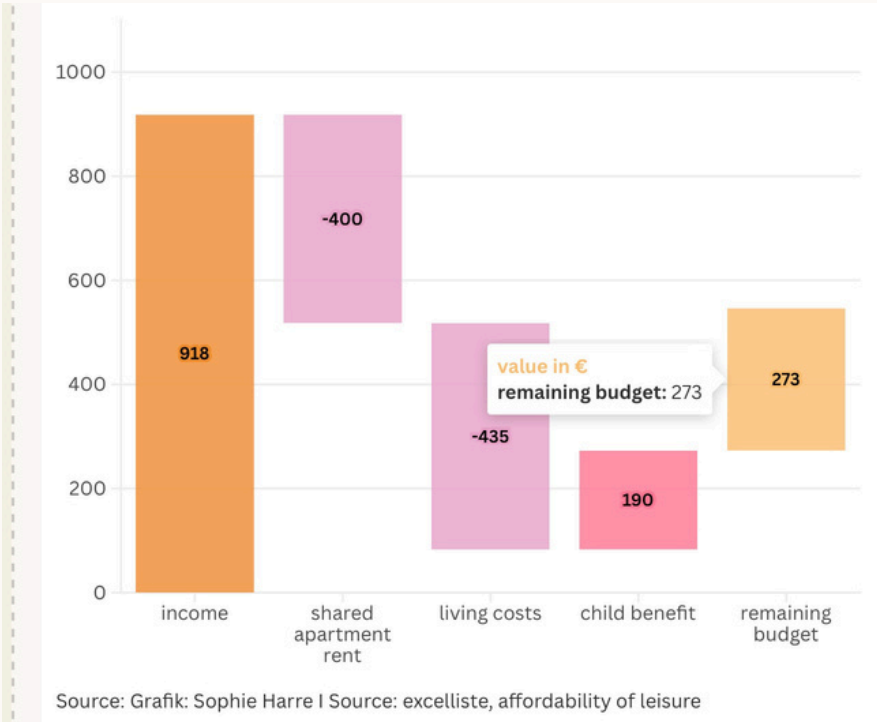


[Get the data](#) • Created with [Datawrapper](#)

VISUALIZATION DESIGN

03 Waterfall Chart

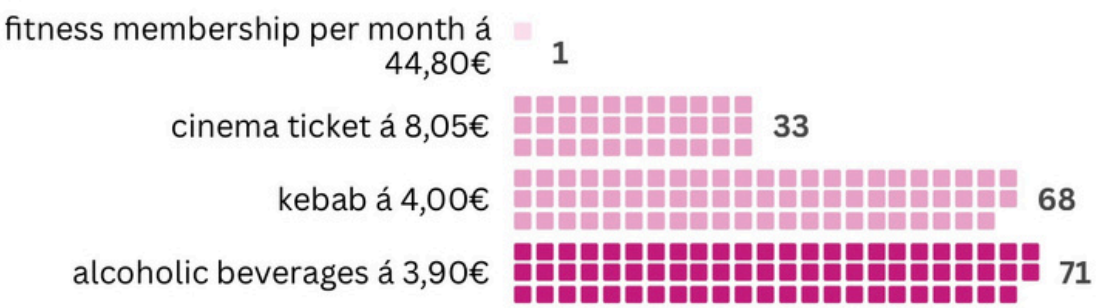
- visualize how the initial value of income is affected by fix costs to reach the final total of remaining budget



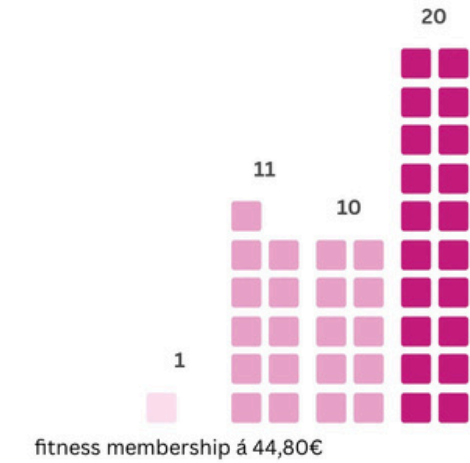
04 Pictogram

- visualize how often leisure activity can be bought

273€ - How many Leisure Activities can Sarah Afford?



What does 273€ of Freedom Monthly Actually look like?



USE OF AI

Use of ChatGPT



- Creating graphics/photos of our main characters Sarah and Emma
- Photos created: 6 different graphics

AI used for creating the website

- Help with creating the Website
 - 1. Creation of a template (based on our storyline) without AI on Canva
 - 2. Use of Claude: building the main structure for our Website as html and css files
 - 3. Implementing our template with long AI prompts with Microsoft CoPilot
 - 4. Adjusting last details and nuances with Microsoft CoPilot and without AI

THANK YOU!

Our Website: Sarah X Emma



IF YOU HAVE ANY QUESTIONS, JUST ASK! :)